

Example 4 Find the equation of the linear function whose contour diagram is in Figure 12.66.

Solution Suppose we start at the origin on the $z = 0$ contour. Moving 2 units in the y -direction takes us to the $z = 6$ contour; so the slope in the y -direction is $\Delta z / \Delta y = 6/2 = 3$. Similarly, a move of 2 in the x -direction from the origin takes us to the $z = 2$ contour, so the slope in the x -direction is $\Delta z / \Delta x = 2/2 = 1$. Since $f(0, 0) = 0$, we have $f(x, y) = x + 3y$.

Exercises and Problems for Section 12.4

Exercises

Exercises 1–2 each contain a partial table of values for a linear function. Fill in the blanks.

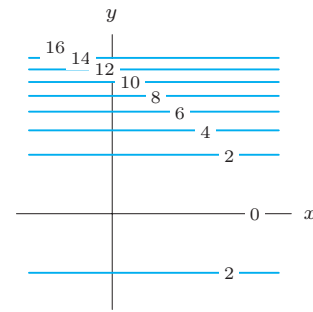
1.

$x \backslash y$	0.0	1.0
0.0		1.0
2.0	3.0	5.0

2.

$x \backslash y$	-1.0	0.0	1.0
2.0	4.0		
3.0		3.0	5.0

8.



In Exercises 3–6, could the tables of values represent a linear function?

3.

	y		
	0	1	2
x	0	0	1
	1	1	0
	2	4	1

4.

	y		
	0	1	2
x	0	10	13
	1	6	9
	2	2	5

9. Find the equation of the linear function $z = c + mx + ny$ whose graph contains the points $(0, 0, 0)$, $(0, 2, -1)$, and $(-3, 0, -4)$.

10. Find the linear function whose graph is the plane through the points $(4, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 2)$.

5.

	y		
	0	1	2
x	0	0	5
	1	2	7
	2	4	9

6.

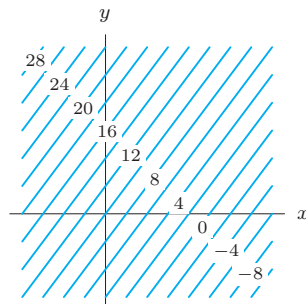
	y		
	0	1	2
x	0	5	7
	1	6	9
	2	7	11

11. Find an equation for the plane containing the line in the xy -plane where $y = 1$, and the line in the xz -plane where $z = 2$.

12. Find the equation of the linear function $z = c + mx + ny$ whose graph intersects the xz -plane in the line $z = 3x + 4$ and intersects the yz -plane in the line $z = y + 4$.

In Exercises 7–8, could the contour diagram represent a linear function?

7.



13. Suppose that z is a linear function of x and y with slope 2 in the x -direction and slope 3 in the y -direction.

- A change of 0.5 in x and -0.2 in y produces what change in z ?
- If $z = 2$ when $x = 5$ and $y = 7$, what is the value of z when $x = 4.9$ and $y = 7.2$?

14. (a) Find a formula for the linear function whose graph is a plane passing through point $(4, 3, -2)$ with slope 5 in the x -direction and slope -3 in the y -direction.
(b) Sketch the contour diagram for this function.

Problems

15. A store sells CDs at one price and DVDs at another price. Figure 12.67 shows the revenue (in dollars) of the music store as a function of the number, c , of CDs and the number, d , of DVDs that it sells. What is the price of a CD? What is the price of a DVD?

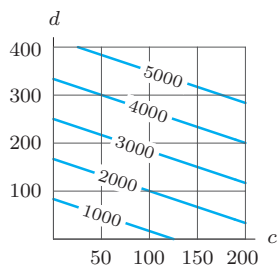


Figure 12.67

16. The charge, C , in dollars, to use an Internet service is a function of m , the number of months of use, and t , the total number of minutes on-line:

$$C = f(m, t) = 35 + 15m + 0.05t.$$

- Is f a linear function?
 - Give units for the coefficients of m and t , and interpret them as charges.
 - Interpret the intercept 35 as a charge.
 - Find $f(3, 800)$ and interpret your answer.
17. A manufacturer makes two products out of two raw materials. Let q_1, q_2 be the quantities sold of the two products, p_1, p_2 their prices, and m_1, m_2 the quantities purchased of the two raw materials. Which of the following functions do you expect to be linear, and why? In each case, assume that all variables except the ones mentioned are held fixed.
- Expenditure on raw materials as a function of m_1 and m_2 .
 - Revenue as a function of q_1 and q_2 .
 - Revenue as a function of p_1 and q_1 .

Problems 18–20 concern Table 12.10, which gives the number of calories burned per minute for someone roller-blading, as a function of the person's weight and speed.⁷

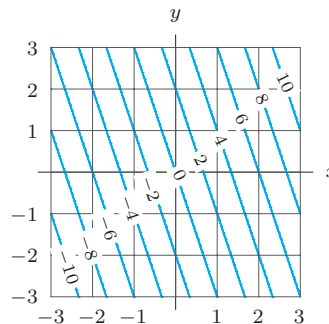
Table 12.10

Calories burned per minute				
Weight	8 mph	9 mph	10 mph	11 mph
120 lbs	4.2	5.8	7.4	8.9
140 lbs	5.1	6.7	8.3	9.9
160 lbs	6.1	7.7	9.2	10.8
180 lbs	7.0	8.6	10.2	11.7
200 lbs	7.9	9.5	11.1	12.6

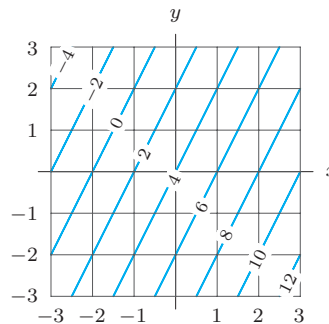
18. Does the data in Table 12.10 look approximately linear? Give a formula for B , the number of calories burned per minute in terms of the weight, w , and the speed, s . Does the formula make sense for all weights or speeds?
19. Who burns more total calories to go 10 miles: A 120 lb person going 10 mph or a 180 lb person going 8 mph? Which of these two people burns more calories per pound for the 10-mile trip?
20. Use Problem 18 to give a formula for P , the number of calories burned per pound, in terms of w and s , for a person weighing w lbs roller-blading 10 miles at s mph.

For Problems 21–22, find possible equations for linear functions with the given contour diagrams.

21.



22.



For Problems 23–24, find an equation for the linear function with the given values.

23.

$x \backslash y$	-1	0	1	2
0	1.5	1	0.5	0
1	3.5	3	2.5	2
2	5.5	5	4.5	4
3	7.5	7	6.5	6

24.

$x \backslash y$	10	20	30	40
100	3	6	9	12
200	2	5	8	11
300	1	4	7	10
400	0	3	6	9

⁷From the August 28, 1994, issue of *Parade Magazine*.

It is difficult to graph a linear function by hand. One method that works if the x , y , and z -intercepts are positive is to plot the intercepts and join them by a triangle as shown in Figure 12.68; this shows the part of the plane in the octant where $x \geq 0$, $y \geq 0$, $z \geq 0$. If the intercepts are not all positive, the same method works if the x , y , and z -axes are drawn from a different perspective. Use this method to graph the linear functions in Problems 25–28.

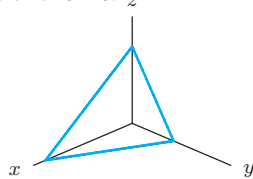


Figure 12.68

25. $z = 2 - 2x + y$ 26. $z = 2 - x - 2y$
 27. $z = 4 + x - 2y$ 28. $z = 6 - 2x - 3y$

29. Let f be the linear function $f(x, y) = c + mx + ny$, where c, m, n are constants and $n \neq 0$.

- (a) Show that all the contours of f are lines of slope $-m/n$.
 (b) For all x and y , show $f(x + n, y - m) = f(x, y)$.
 (c) Explain the relation between parts (a) and (b).

Problems 30–31 refer to the linear function $z = f(x, y)$ whose values are in Table 12.11.

Table 12.11

		y				
		4	6	8	10	12
x	5	3	6	9	12	15
	10	7	10	13	16	19
	15	11	14	17	20	23
	20	15	18	21	24	27
	25	19	22	25	28	31

30. Each column of Table 12.11 is linear with the same slope, $m = \Delta z / \Delta x = 4/5$. Each row is linear with the same slope, $n = \Delta z / \Delta y = 3/2$. We now investigate the slope obtained by moving through the table along lines that are neither rows nor columns.

- (a) Move down the diagonal of the table from the upper left corner ($z = 3$) to the lower right corner ($z = 31$). What do you notice about the changes in z ? Now move diagonally from $z = 6$ to $z = 27$. What do you notice about the changes in z now?

- (b) Move in the table along a line right one step, up two steps from $z = 19$ to $z = 9$. Then move in the same direction from $z = 22$ to $z = 12$. What do you notice about the changes in z ?

- (c) Show that $\Delta z = m\Delta x + n\Delta y$. Use this to explain what you observed in parts (a) and (b).

31. If we hold y fixed, that is we keep $\Delta y = 0$, and step in the positive x -direction, we get the x -slope, m . If instead we keep $\Delta x = 0$ and step in the positive y -direction, we get the y -slope, n . Fix a step in which neither $\Delta x = 0$ nor $\Delta y = 0$. The slope in the $\Delta x, \Delta y$ direction is

$$\begin{aligned} \text{Slope} &= \frac{\text{Rise}}{\text{Run}} = \frac{\Delta z}{\text{Length of step}} \\ &= \frac{\Delta z}{\sqrt{(\Delta x)^2 + (\Delta y)^2}}. \end{aligned}$$

- (a) Compute the slopes for the linear function in Table 12.11 in the direction of $\Delta x = 5, \Delta y = 2$.

- (b) Compute the slopes for the linear function in Table 12.11 in the direction of $\Delta x = -10, \Delta y = 2$.

32. For the contour diagrams (I)–(IV) on $-2 \leq x, y \leq 2$, pick the corresponding function.

$$f(x, y) = 2x + 3y + 10$$

$$k(x, y) = -2x + 3y + 12$$

$$g(x, y) = 2x + 3y + 60$$

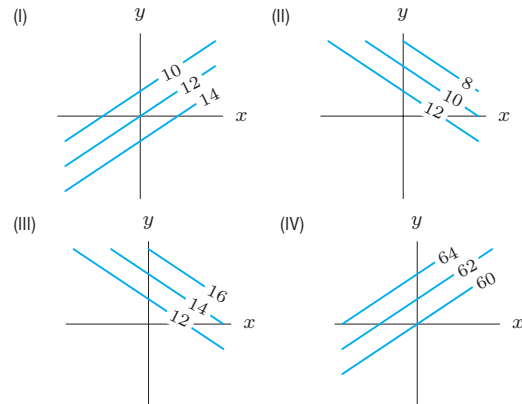
$$m(x, y) = -2x + 3y + 60$$

$$h(x, y) = 2x - 3y + 12$$

$$n(x, y) = -2x - 3y + 14$$

$$j(x, y) = 2x - 3y + 60$$

$$p(x, y) = -2x - 3y + 60$$



Strengthen Your Understanding

In Problems 33–34, explain what is wrong with the statement.

33. If the contours of f are all parallel lines, then f is linear.

34. A function $f(x, y)$ with linear cross-sections for x fixed and linear cross-sections for y fixed is a linear function.

In Problems 35–36, give an example of:

35. A table of values, with three rows and three columns, for a nonlinear function that is linear in each row and in each column.
36. A linear function whose contours are lines with slope 2.

Are the statements in Problems 37–48 true or false? Give reasons for your answer.

37. The planes $z = 3 + 2x + 4y$ and $z = 5 + 2x + 4y$ intersect.
38. The function represented in Table 12.12 is linear.

Table 12.12

$u \backslash v$	1.1	1.2	1.3	1.4
3.2	11.06	12.06	13.06	14.06
3.4	11.75	12.82	13.89	14.96
3.6	12.44	13.58	14.72	15.86
3.8	13.13	14.34	15.55	16.76
4.0	13.82	15.10	16.38	17.66

39. Contours of $f(x, y) = 3x + 2y$ are lines with slope 3.
40. If f is linear, then the contours of f are parallel lines.
41. If $f(0, 0) = 1$, $f(0, 1) = 4$, $f(0, 3) = 5$, then f cannot be linear.
42. The graph of a linear function is always a plane.
43. The cross-section $x = c$ of a linear function $f(x, y)$ is always a line.
44. There is no linear function $f(x, y)$ with a graph parallel to the xy -plane.
45. There is no linear function $f(x, y)$ with a graph parallel to the xz -plane.
46. A linear function $f(x, y) = 2x + 3y - 5$, has exactly one point (a, b) satisfying $f(a, b) = 0$.
47. In a table of values of a linear function, the columns have the same slope as the rows.
48. There is exactly one linear function $f(x, y)$ whose $f = 0$ contour is $y = 2x + 1$.

12.5 FUNCTIONS OF THREE VARIABLES

In applications of calculus, functions of any number of variables can arise. The density of matter in the universe is a function of three variables, since it takes three numbers to specify a point in space. Models of the US economy often use functions of ten or more variables. We need to be able to apply calculus to functions of arbitrarily many variables.

One difficulty with functions of more than two variables is that it is hard to visualize them. The graph of a function of one variable is a curve in 2-space, the graph of a function of two variables is a surface in 3-space, so the graph of a function of three variables would be a solid in 4-space. Since we can't easily visualize 4-space, we won't use the graphs of functions of three variables. On the other hand, it is possible to draw contour diagrams for functions of three variables, only now the contours are surfaces in 3-space.

Representing a Function of Three Variables Using a Family of Level Surfaces

A function of two variables, $f(x, y)$, can be represented by a family of level curves of the form $f(x, y) = c$ for various values of the constant, c .

A **level surface**, or **level set** of a function of three variables, $f(x, y, z)$, is a surface of the form $f(x, y, z) = c$, where c is a constant. The function f can be represented by the family of level surfaces obtained by allowing c to vary.

The value of the function, f , is constant on each level surface.

Example 1 The temperature, in $^{\circ}\text{C}$, at a point (x, y, z) is given by $T = f(x, y, z) = x^2 + y^2 + z^2$. What do the level surfaces of the function f look like and what do they mean in terms of temperature?

Solution The level surface corresponding to $T = 100$ is the set of all points where the temperature is 100°C . That is, where $f(x, y, z) = 100$, so

$$x^2 + y^2 + z^2 = 100.$$